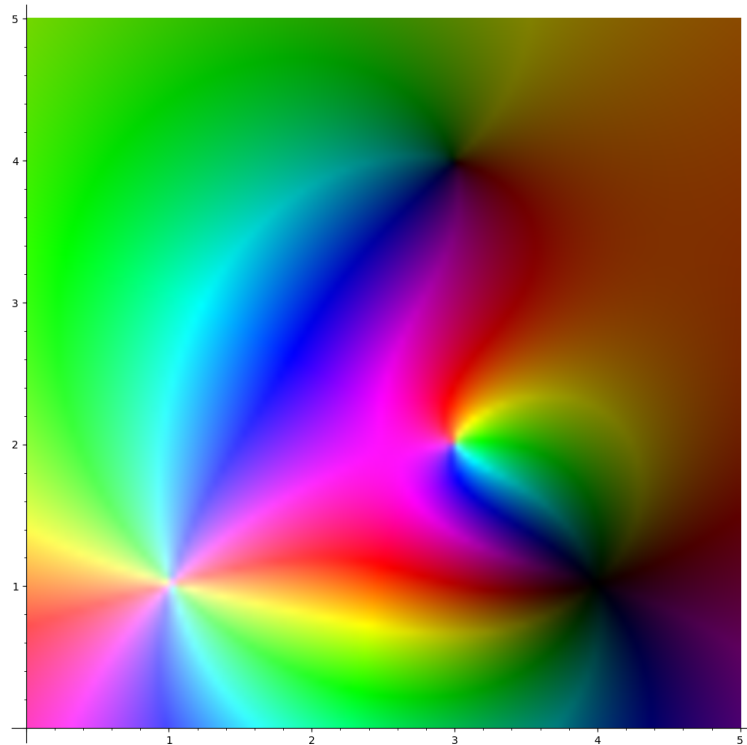


Complex Variable - Midterm Mock Exam

考試時間 Test time 10:30:00 am to 12:00:00 pm; 1.5 hours. 以教室投影機時鐘為準 Using the projected clock.
用空白紙張作答 Answer on blank paper. 每題十分 Ten points per problem. 全對才給分 No partial credit. 可以帶
991 You can use fx-991 or anything weaker. 舉手問問題 Raise your hand to ask questions.

[1] Circle all the roots. If it is a double root, circle twice, and so on. For a pole, use squares instead of circles. Mark the positive real line with solid lines. Mark the negative real line with dotted lines. (Don't answer here; draw the result on your answer sheet.)



[2] What maps the strip $1 < x - y < 2$ to the fan $1 < \arg z < 2$?

[3] Find the Taylor expansion of $\sin(\cos(z))$ at $z = 0$ up to and including the z^6 term.

[4] Assume that the Taylor expansion of h exists. Use Taylor to show that the line integral along a small square of side length ε is $O(\varepsilon^3)$. (Don't use Green or Cauchy or so; as we are trying to "verify" it.)

[5] Let S be a subset of $\mathbb{C}$. We know that, for any converging sequence of points taken from S , the limit is still in S . Prove that for any point in the complement of S , there is an open ball around that point that does not touch S .

[6] Find (and prove) the radius of convergence of the following series $a_n := (10 + \exp(in))^n$.

[7] The Gamma function $\Gamma(z)$ is defined as

$$\int_0^\infty t^{z-1} \exp(-t) dt$$

for all z with positive real part. Prove that $\Gamma(z+1) = z\Gamma(z)$ for all such z .

[8] Integrate, numerically, $\exp(z^2)$ from 0 to $\pi/2 + i$ along the straight line. Your score is the largest number k in $0 \sim 10$ such that the additive error is $< 2^{-k}$.

[9] Find the winding number of $10 \exp(it) + 9 \exp(8it)$ for t from 0 to 2π w.r.t, $4 + 2i$.

[10] $x^5 - 40x^3 + 3000i$ has one and only one root in the first quadrant. Find the unit square that contains it. Hint: start with a 16×16 square and count the number of roots by integrating $f'/2\pi i f$. If yes, cut it into four 8×8 smaller squares and repeat.